MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CAPACITOR, PROGRAMMABLE, ANALOG, KEY CODE GGD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic programmable capacitor module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.15 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GGD.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 70 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned A1 through A20.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/179-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

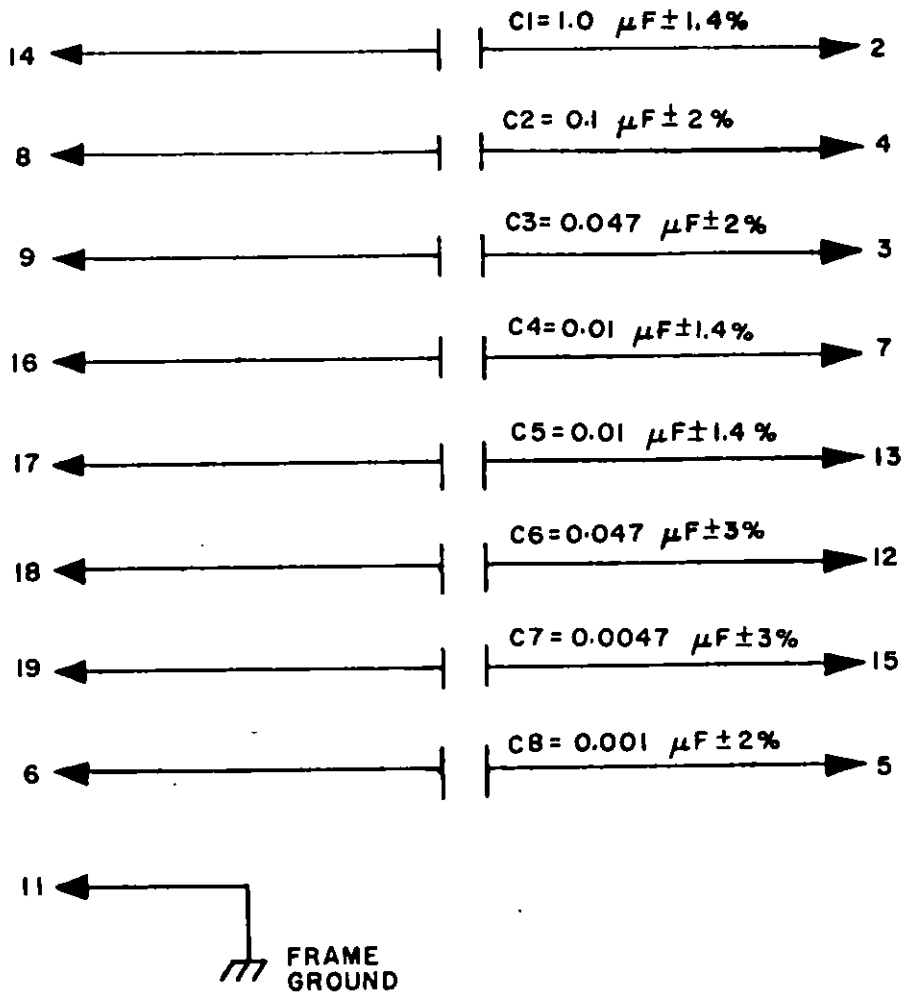
* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through E shall be performed in any order with the following exceptions. If test condition A is part of a sequence of tests, the order of testing shall be as follows: A where applicable, followed by the remaining tests in the sequence.

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NOTES:
1. All capacitors are 30 working volts dc, nonpolarized.
2. All capacitance values are for a class I environment.
3. Pins not used = 1, 10, and 20.

FIGURE 1. GGD module function diagram, programmable capacitor.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-2/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	Isolation	II	X	100	∞	10	∞	100	∞	10	∞	MΩ
B1-B16	DC leakage current	III	X	0	3	0	30	0	3	0	30	μA
C1	Capacitance	IV	X	0.9878	1.0122	0.9859	1.0141	0.9859	1.0141	0.9859	1.0141	μF
C2	Capacitance	IV	X	0.09878	0.10122	0.0980	0.1020	0.0980	0.1020	0.0980	0.1020	μF
C3	Capacitance	IV	X	0.04643	0.04757	0.04606	0.04794	0.04606	0.04794	0.04606	0.04794	μF
C4,C5	Capacitance	IV	X	0.00988	0.01012	0.00986	0.01014	0.00986	0.01014	0.00986	0.01014	μF
C6	Capacitance	IV	X	0.04633	0.04766	0.04559	0.04841	0.04559	0.04841	0.04559	0.04841	μF
C7	Capacitance	IV	X	0.004606	0.004794	0.004559	0.004841	0.004559	0.004841	0.004559	0.004841	μF
C8	Capacitance	IV	X	0.000983	0.001017	0.000980	0.001020	0.000980	0.001020	0.000980	0.001020	μF
D1-D8	Dissipation factor	V	X	0	1	0	1	0	1	0	1	%
E	Continuity	VI	S	0	1	0	1	0	1	0	1	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

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4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through E in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through D shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

* 4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A through E shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through E shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number MIL-M-28787/179-1 supersedes part number M28787/179. The two parts are interchangeable.

* 6.4 Module function. This module contains eight independent capacitors. Capacitor values, tolerances, and voltage ratings are specified on figure 1.

* 6.4.1 System application. The voltage rating of the capacitors shall be a minimum of 50 V dc. However, 30 V dc is the maximum voltage that should be applied in system application. If greater than 30 V dc is applied to any capacitor, the specified failure rate is no longer applicable.

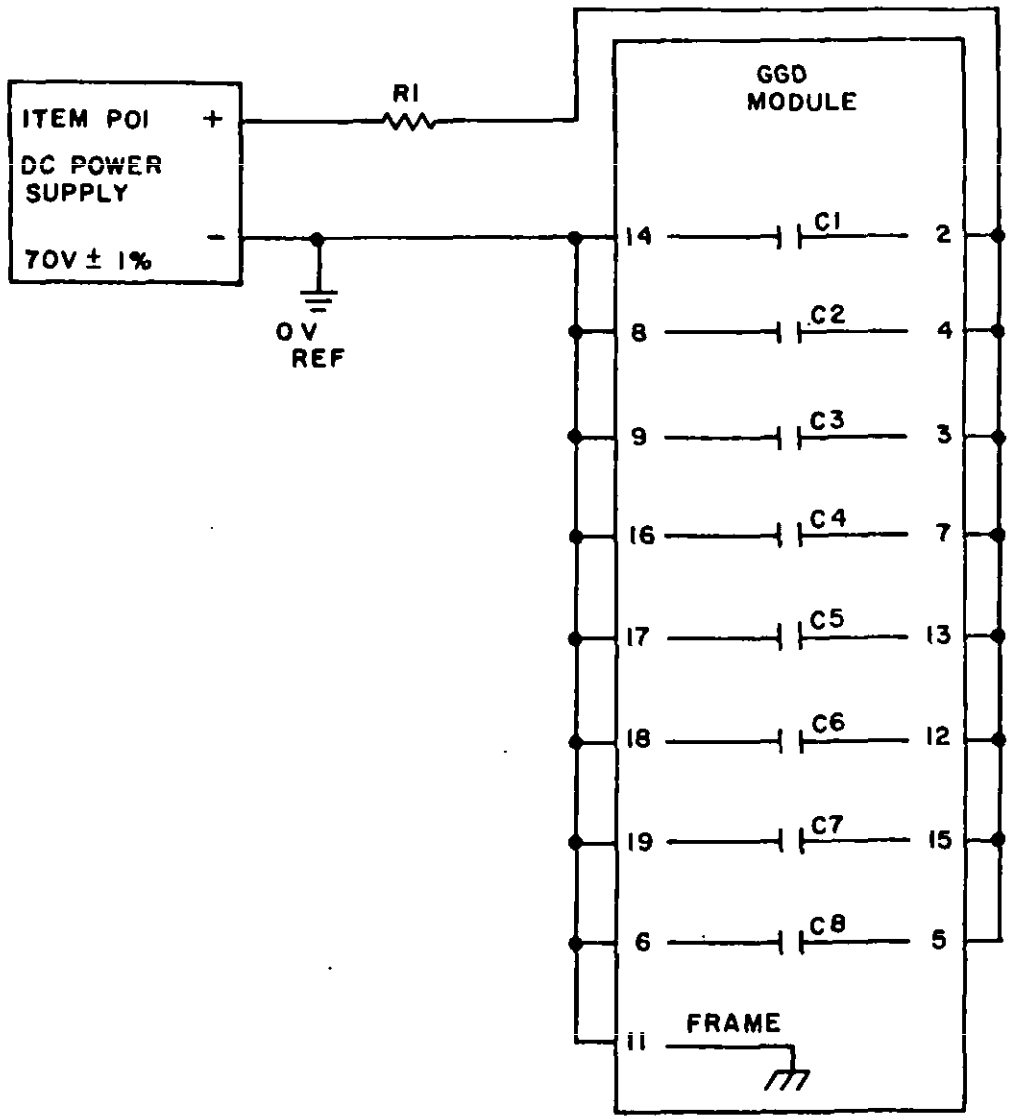
6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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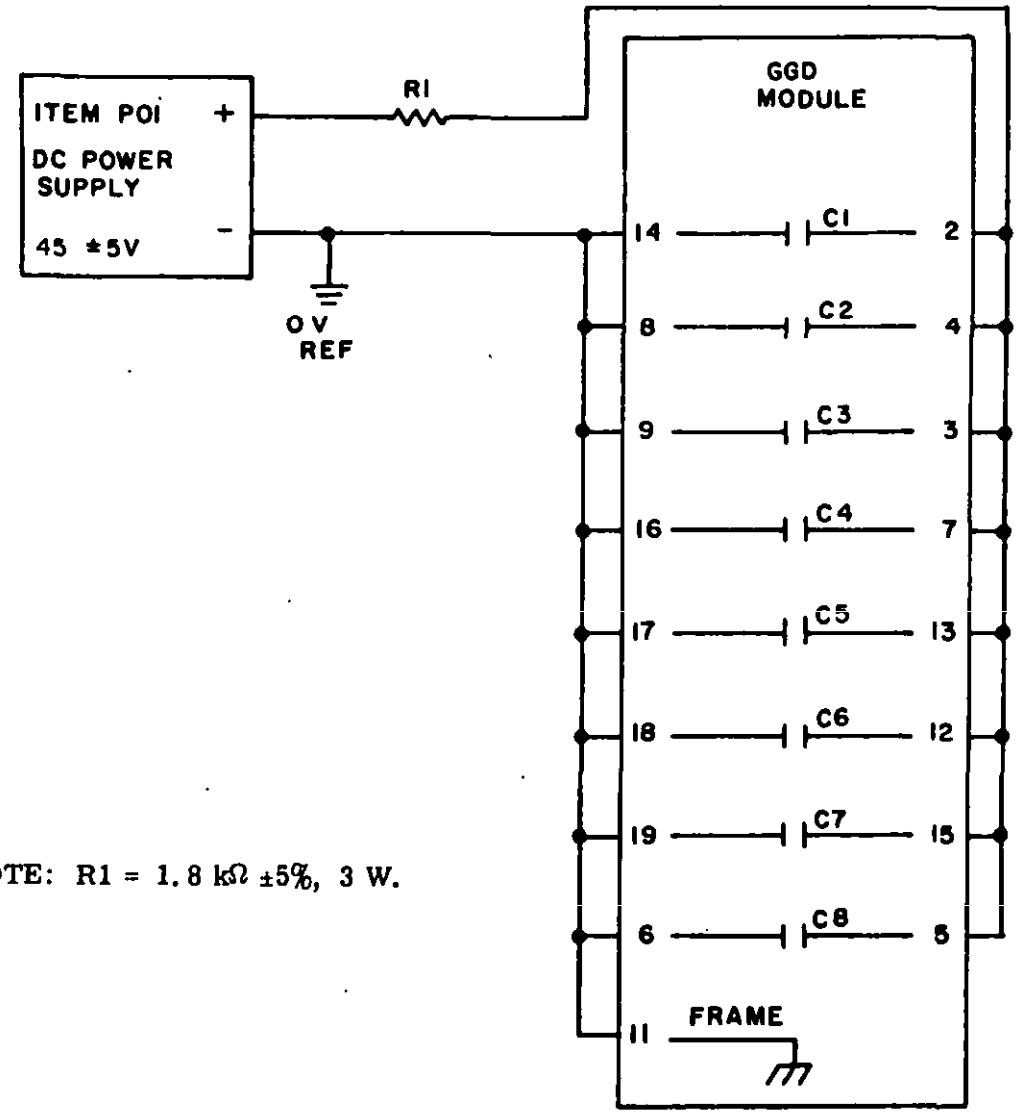
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NOTES:
1. R1 = 1.8 kΩ ±5%, 3 W.
2. For the first 250 hours, use the test setup shown. For the second 250 hours, reverse the polarity of voltage applied to the capacitors.
3. The item POI voltage shall be up to 140 percent of the module capacitors rated voltage.

* FIGURE 2. Operating temperature, component temperature, and life test circuit.

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NOTE: R1 = 1.8 kΩ ±5%, 3 W.

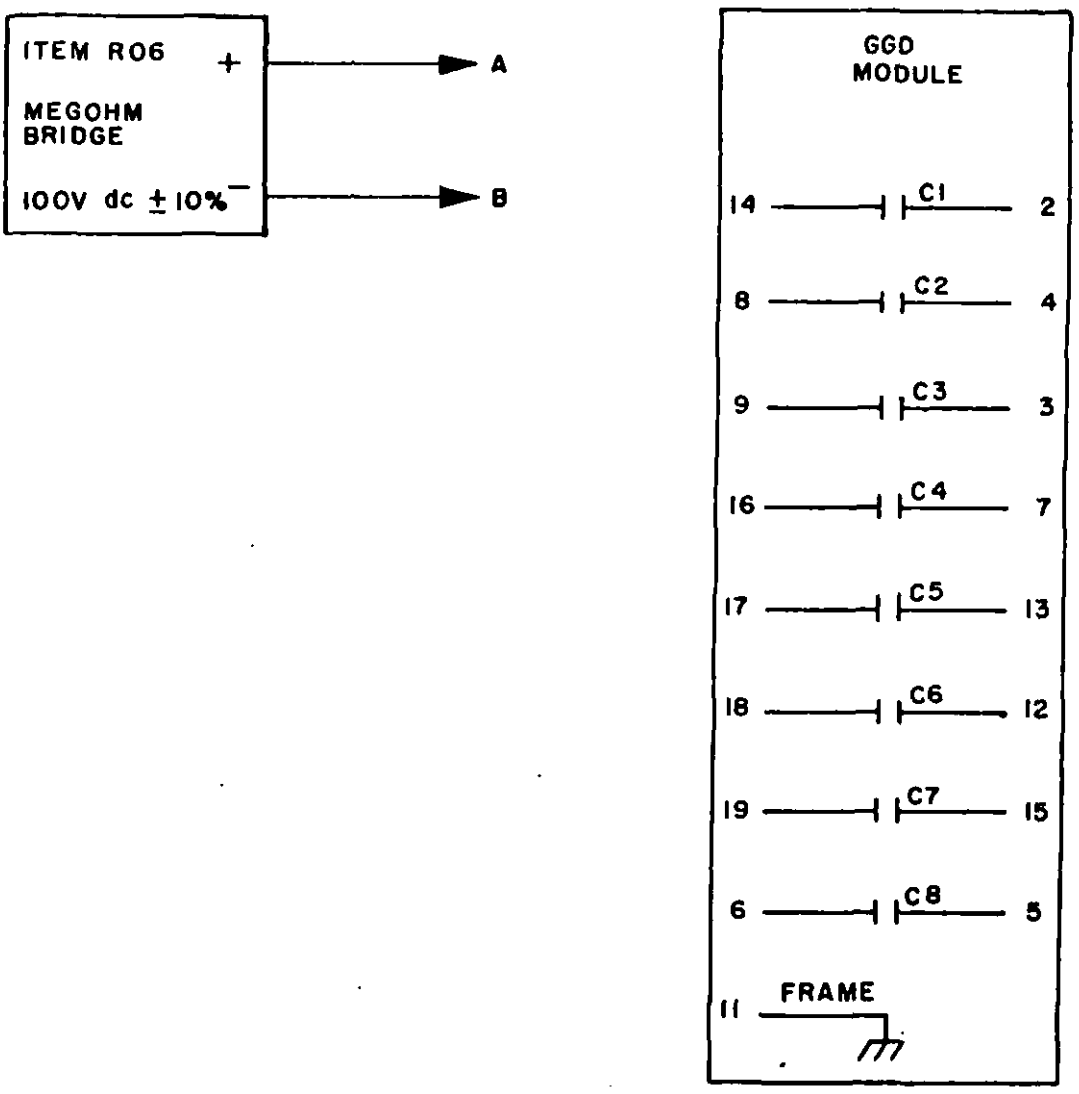
FIGURE 3. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, module pins to frame pin	1-10,12-20	11	100	∞	MΩ	4
A2	Isolation, used to unused pins	2-9,11-19	1,10,20	"	"	"	"
A3	Isolation, C1 circuit to other circuits	3-9,11-13,15-19	14,2	"	"	"	"
A4	Isolation, C2 circuit to other circuits	2,3,5-7,9,11-19	8,4	"	"	"	"
A5	Isolation, C3 circuit to other circuits	2,4-8,11-19	9,3	"	"	"	"
A6	Isolation, C4 circuit to other circuits	2-6,8,9,11-15, 17-19	16,7	"	"	"	"
A7	Isolation, C5 circuit to other circuits	2-9,11,12, 14-16,18,19	17,13	"	"	"	"
A8	Isolation, C6 circuit to other circuits	2-9,11,13-17,19	18,12	"	"	"	"
A9	Isolation, C7 circuit to other circuits	2-9,11-14,16-18	19,15	"	"	"	"
A10	Isolation, C8 circuit to other circuits	2-4,7-9,11-19	6,5	"	"	"	"
A11	Isolation, unused to unused pins	10,20	1	"	"	"	"
A12	Isolation, unused to unused pins	10	20	"	"	"	"

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to discharge before connecting or disconnecting the test leads.

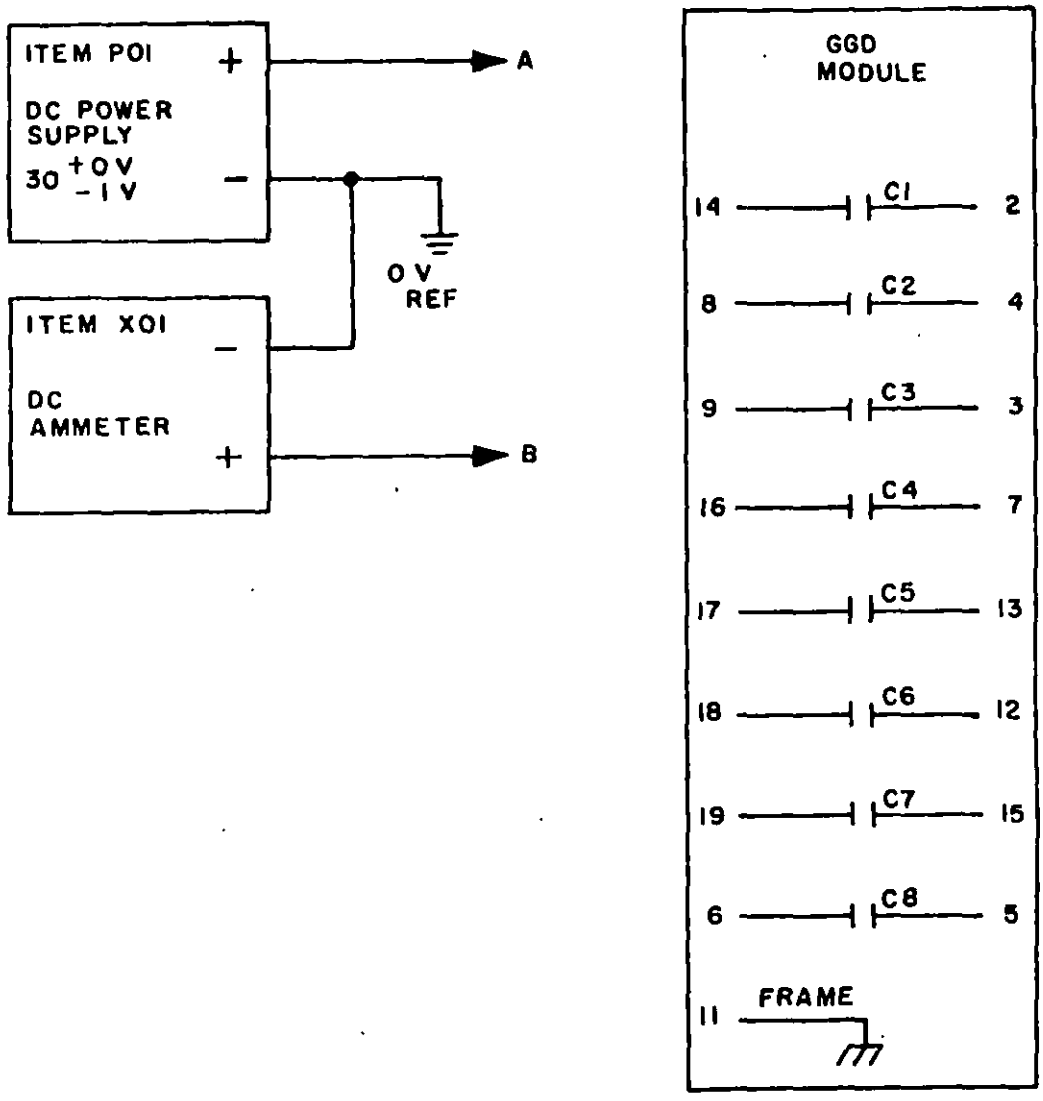
FIGURE 4. Isolation test circuit for table II.

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*TABLE III. DC leakage current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	C1 DC leakage current	14	2	0	3	µA	5
B2	C2 "	8	4	"	"	"	"
B3	C3 "	9	3	"	"	"	"
B4	C4 "	16	7	"	"	"	"
B5	C5 "	17	13	"	"	"	"
B6	C6 "	18	12	"	"	"	"
B7	C7 "	19	15	"	"	"	"
B8	C8 "	6	5	"	"	"	"
B9	C1 "	2	14	"	"	"	"
B10	C2 "	4	8	"	"	"	"
B11	C3 "	3	9	"	"	"	"
B12	C4 "	7	16	"	"	"	"
B13	C5 "	13	17	"	"	"	"
B14	C6 "	12	18	"	"	"	"
B15	C7 "	15	19	"	"	"	"
B16	C8 "	5	6	"	"	"	"

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CAUTION: Do not isolate the chassis of item X01 from earth ground.

NOTE: The unit under test conforms to requirements if leakage current drops below the limit indicated in table III in 5 minutes or less.

* FIGURE 5. DC leakage current test circuit for table III.

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TABLE IV. Capacitance test.

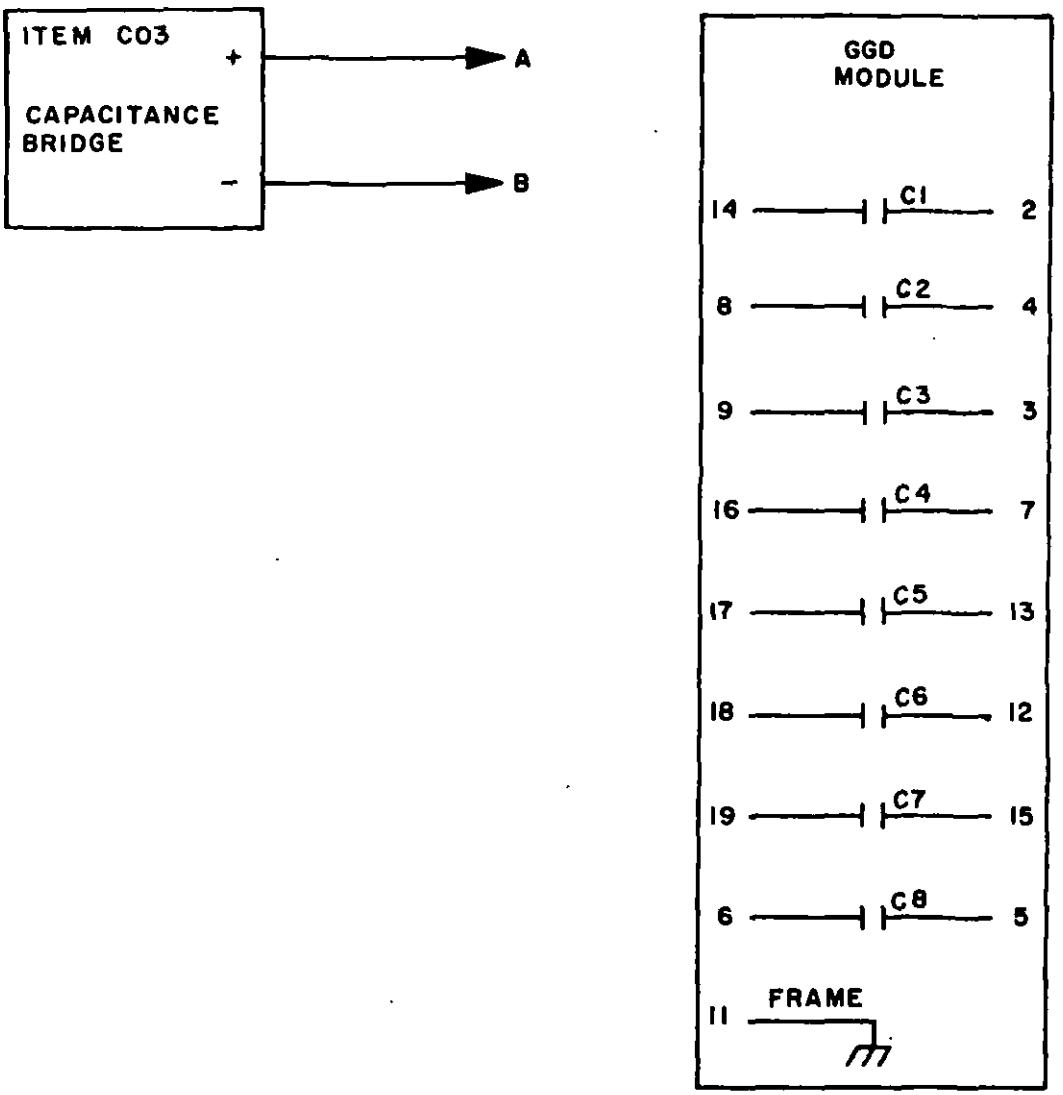
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	C1 Capacitance	14	2	0.9878	1.0122	µF	6
C2	C2 "	8	4	0.09878	0.10122	"	"
C3	C3 "	9	3	0.04643	0.04757	"	"
C4	C4 "	16	7	0.00988	0.01012	"	"
C5	C5 "	17	13	0.00988	0.01012	"	"
C6	C6 "	18	12	0.04633	0.04766	"	"
C7	C7 "	19	15	0.004606	0.004794	"	"
C8	C8 "	6	5	0.000983	0.001017	"	"

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TABLE V. Dissipation factor test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	C1 Dissipation factor	14	2	0	1	%	6
D2	C2 "	8	4	"	"	"	"
D3	C3 "	9	3	"	"	"	"
D4	C4 "	16	7	"	"	"	"
D5	C5 "	17	13	"	"	"	"
D6	C6 "	18	12	"	"	"	"
D7	C7 "	19	15	"	"	"	"
D8	C8 "	6	5	"	"	"	"

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pins under test. Stray capacitance added to other pins may affect the readings.
3. Item CO3 setup:
   a. Frequency = 1 kHz ±100 Hz.
   b. V_{pk} = 1 V maximum.

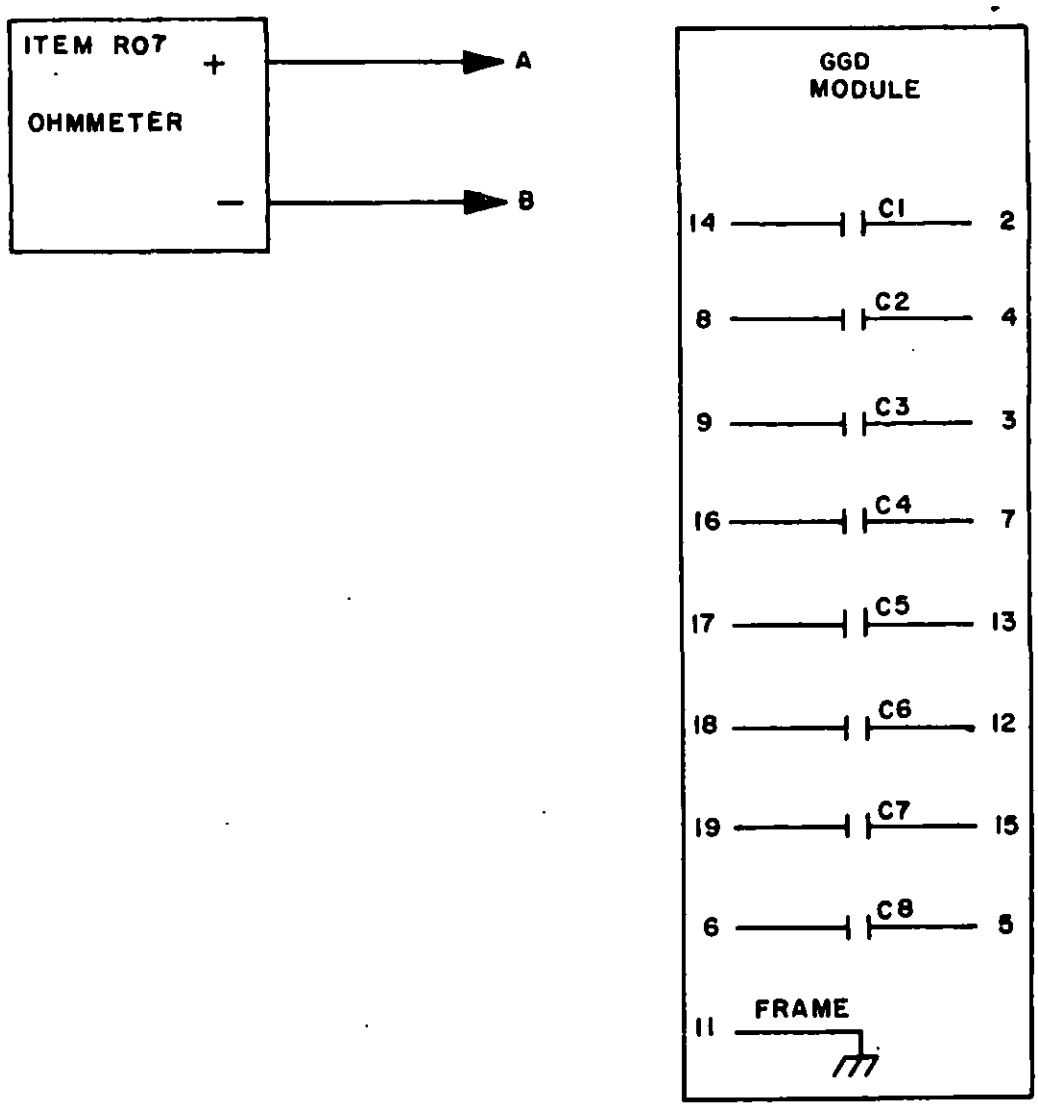
* FIGURE 6. Capacitance and dissipation factor test circuit for tables IV and V.

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TABLE VI. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E	Continuity	11	Frame	0	1	Ω	7

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NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 7. Continuity test circuit for table VI.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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MIL-M-28787/179B(EC	Modules, Standard Electronic, Capacitor, Programmable,

3a. NAME OF SUBMITTING ORGANIZATION Analog, Key Code GGD	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
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